

1.

'In tackling environmental problems there is not really a straight choice between perfectly performing markets on the one hand, and wonderfully wise governments on the other.'
(**Extract C**, lines 18 – 19).

Using the data and your knowledge of economics, assess the relative importance of market forces and government intervention in encouraging businesses to behave responsibly towards the environment.

[25 marks]

2.

The table below shows the marginal private and external benefits and the marginal private and external costs of two products at their free market equilibrium level of output.

	Product X (£)	Product Y (£)
Marginal private benefit	200	70
Marginal external benefit	50	40
Marginal private cost	200	70
Marginal external cost	90	30

To improve the allocation of resources, what should the government do?

A Tax product X and subsidise product Y

☐

B Tax product X and tax product Y

☐

C Subsidise product X and tax product Y

☐

D Subsidise product X and subsidise product Y

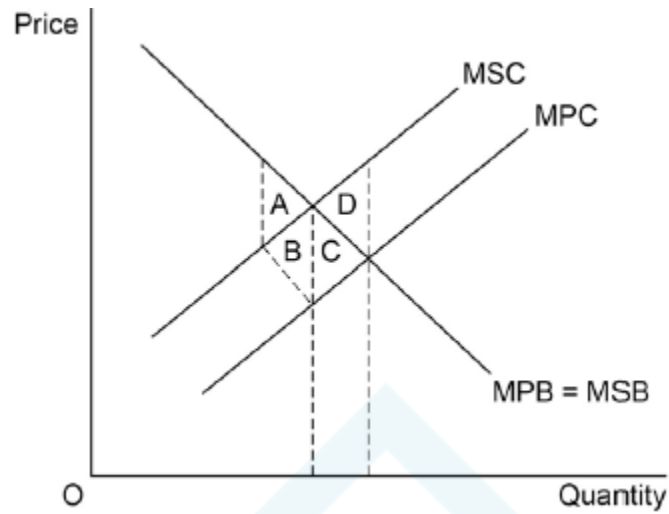
☐

[1 mark]



3.

The diagram below shows the marginal private benefit and marginal social benefit (MPB and MSB) curves and marginal private cost and marginal social cost (MPC and MSC) curves for Good X.



Which triangle, **A**, **B**, **C** or **D**, represents the welfare loss if output is at the private optimum rather than the social optimum?

A ☐

B ☐

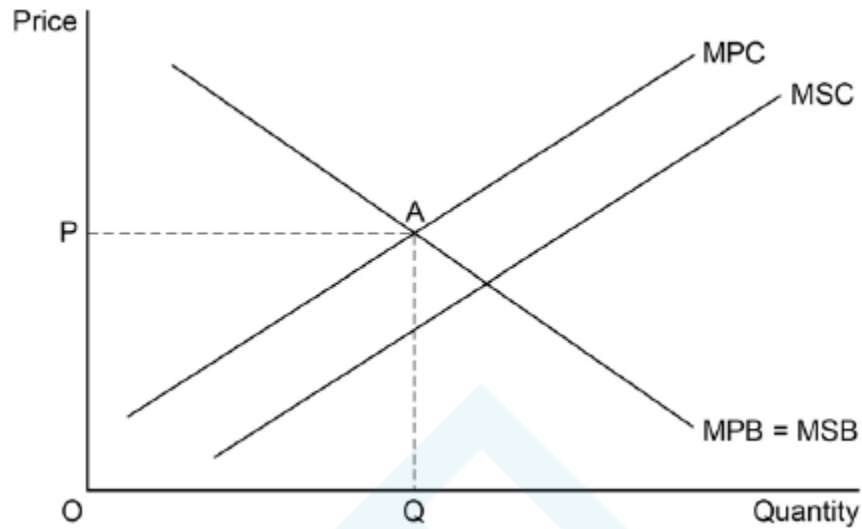
C ☐

D ☐

[1 mark]

4.

The diagram below shows the marginal private benefit and the marginal social benefit (MPB and MSB) curves and the marginal private cost and marginal social cost (MPC and MSC) curves for good X.



Assuming the free market equilibrium is at point A, what action taken by the government is most likely to lead to output being produced at the socially optimal level?

- A Increase in unit taxes on good X ☐
- B Subsidies to the producers of good X ☐
- C Regulation to reduce consumption of good X ☐
- D Issue of permits designed to reduce output of good X ☐

[1 mark]

5.

Pollution and Climate Change Extract F (lines 9–10) states ‘Some major UK businesses appear happier with permit trading than with alternative policies such as quotas on emissions or the taxation of fuel’.

Using the data in the extracts and your knowledge of economics, evaluate the possible impact on UK businesses of policies to reduce pollution.

[25 marks]



6.

THE EUROPEAN UNION CONTEXT**Extract E: Housing tenure and externalities**

Most Spaniards and Estonians live in flats, while most people in the UK and Ireland live in semi-detached houses. Housing markets are made up of different sectors or 'tenures' which describe the legal basis under which houses are occupied. Most of the population of the EU live in owner-occupied houses. Social housing is built or funded by the public sector and, in the past, the UK has been almost unique in Europe in providing such houses mainly through local councils. In France, this is done by not-for-profit organisations, similar to UK housing associations. About two-thirds of tenants across the EU pay a market price rent, while the rest receive some form of subsidy. In the UK, the concept of social housing has become complicated by the large number of private landlords whose tenants are receiving welfare benefits. While the UK government has moved away from rent controls, it has attempted to influence rents indirectly by capping housing benefits for low income tenants.

While housing has many of the characteristics of a merit good, externalities from housing can be positive or negative. Bad housing is often associated with crime, ill-health and other problems, and the opposite is true of good housing. Tenure can also be linked to externalities, for example, if poorly designed social housing, such as an inner-city block of flats, is compared with high-quality private housing in a prosperous area. Much depends on the quality of the build, the surrounding social conditions and the economic infrastructure.

Source: News reports, 2014

7.

Pollution and Climate Change Extract E (lines 18-20) states 'They argue that the pollution and depletion of communal resources such as clean air, water reserves and fish stocks are made worse by a lack of property rights'.

With the help of a diagram, explain how the absence of property rights can worsen the environmental market failures of **both** pollution **and** the rapid depletion of natural resources.

[9 marks]

8.

Pollution and climate change

Extract E: Climate change

In 2004, Russia signed up to the Kyoto Protocol, an international agreement on policies to manage climate change. This committed Russia to a United Nations programme to reduce emissions of greenhouse gases. In 2010, Russia's emissions were as low as their 1990 levels, but this was less to do with environmental policy and more to do with the collapse of heavy industry following the fall of the Soviet Union. A large contribution towards the UK's Kyoto targets also came from deindustrialisation. However, China, like the USA, did not sign up to the Kyoto Protocol. 1

International action to control carbon emissions is necessary. One policy is known as 'carbon trading'. Countries agree overall emissions limits and try to create an international market for permits to pollute. The idea is to put a money value on emissions and, over time, to create incentives for businesses to invest in less-polluting technology. As well as the large compulsory 'compliance' market for carbon credits, there is also a smaller voluntary market for 'carbon offsets'. 5

For example, an organisation can be paid to plant trees to allegedly offset the damage that an airline journey does to the ozone layer. Some companies are using these offsets, together with greater energy efficiency and reduced polluting activities, to claim to be 'carbon neutral'. Some environmentalists and economists are suspicious of offset schemes and claim, for instance, that offset schemes have hidden environmental costs. They argue that the pollution and depletion of communal resources such as clean air, water reserves and fish stocks are made worse by a lack of property rights. 10

Source: news reports, 2010

9.

Pollution and climate change

Extract F: Issues over carbon trading

Critics say that carbon trading amounts to 'privatisation of the atmosphere', with property rights in this vast new market being grabbed by speculators or by those who already pollute the most. They also claim that activities such as buying credits from Russia are an 'accounting fraud' because they do not necessarily arise from energy efficiency but from the collapse of heavy industry. There is concern that carbon trading is a further blow to manufacturing and that service industries are treated more leniently. If the UK relies on manufactured imports, it can be argued that factories located in China are actually creating UK emissions. 1

Some major UK businesses appear happier with permit trading than with alternative policies such as quotas on emissions or the taxation of fuel. There is a suspicion that carbon trading might mislead the public into believing that pollution from airlines and other industries is no longer damaging. Governments are accused of being overgenerous in allotting permits to industries that they see as nationally important, such as car manufacturing and electricity generation. While the UK might reduce its domestic demand for carbon, it continues to contribute to the world supply of fossil fuels through activities such as oil prospecting and opencast mining. Russia, China and less-developed countries with coal reserves show few signs of reducing their output of this form of carbon. 5

Source: news reports, 2010



10.

It is generally accepted that we need to adopt new ways of charging for the use of our roads. This is partly because of worsening congestion, but is also due to the reduction in revenue generated from taxes on fuels as a result of more fuel-efficient vehicles.

Assess the view that a system of road-pricing is the best way to tackle the problem of worsening traffic congestion in the UK.

[25 marks]

11.

It is generally accepted that we need to adopt new ways of charging for the use of our roads. This is partly because of worsening congestion, but is also due to the reduction in revenue generated from taxes on fuels as a result of more fuel-efficient vehicles.

Explain why, in the absence of government intervention, too many journeys are likely to be made by road and too few by rail.

[15 marks]

12.

'Most consumers know, more or less, when and why they are likely to want to replace a TV set or mobile phone. Nobody really knows in advance when or why they might need to go into hospital for a new knee-cap or hip joint.'

Discuss whether economic welfare would necessarily be increased if the provision of merit goods and public goods were always left to governments without ever involving markets.

[25 marks]

13.

'The UK government would be happy with state control of British railways "as long as it's not the British state"' (**Extract C**, lines 19 – 21).

Evaluate, using the data and your knowledge of economics, the case for and against the UK government privatising public services while allowing foreign ownership of these services.

[25 marks]

14.

'Most consumers know, more or less, when and why they are likely to want to replace a TV set or mobile phone. Nobody really knows in advance when or why they might need to go into hospital for a new knee-cap or hip joint.'

Explain why imperfect information might lead to market failure in the case of merit goods and demerit goods.

[15 marks]

15.

In 2018, the UK Government will impose a new tax on sugary drinks, aimed at tackling obesity. Behavioural economists believe that taxation alone is unlikely to be very effective in reducing consumption. By 2050, obesity is predicted to affect 60% of adult men, 50% of adult women and 25% of children. It is associated with a range of health problems including type 2 diabetes, cardiovascular disease and cancer.

Evaluate the view that imposing a tax is the most effective government policy for reducing the market failures arising from overconsumption of unhealthy food and drink.

[25 marks]

16.

'A third runway at Heathrow is essential for the success of the economy of London and the UK.'

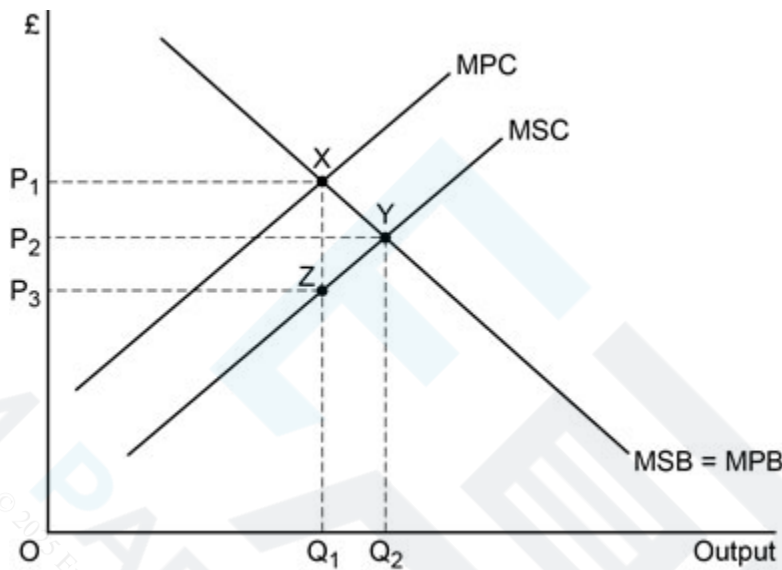
'A third runway at Heathrow would be an environmental disaster and cause huge market failure.'

Explain how an investment project, such as airport expansion, can be commercially successful and yet produce market failure.

[15 marks]

17.

The diagram below shows the marginal private and social benefit (MPB and MSB) curves and the marginal private and social cost (MPC and MSC) curves in a market where there are positive externalities in production.



At the free market equilibrium, market failure occurs. Which one of the following is correct?

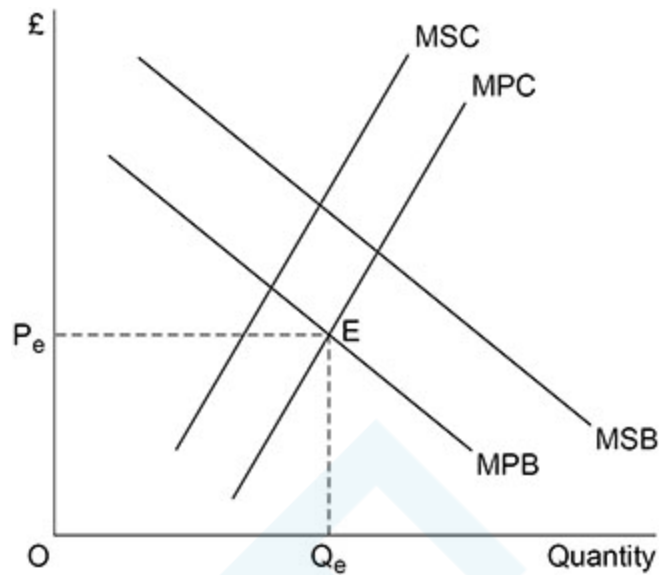
- A As a result of this market failure, consumer surplus is reduced by ZXY
- B As a result of this market failure, there is a deadweight loss to society of P_3P_1XZ
- C P_1 minus P_3 is the subsidy per unit that is required to correct the market failure and achieve the social optimum
- D The diagonal distance XY shows the amount by which marginal private cost exceeds marginal social cost at OQ_2

☐
☐
☐
☐

[1 mark]

18.

The diagram below shows the marginal private benefit and marginal social benefit (MPB and MSB) curves and the marginal private cost and marginal social cost (MPC and MSC) curves in the market for a good.



The diagram represents a good with

- A negative externalities in production and consumption. ☐
- B negative externalities in production and positive externalities in consumption. ☐
- C positive externalities in production and consumption. ☐
- D positive externalities in production and negative externalities in consumption. ☐

[1 mark]

19.

According to Public Health England, poor air quality is the largest environmental risk to public health in the UK. In particular, diesel cars are estimated to be responsible for 40 000 premature deaths each year. In response, the Government announced in 2017 that it would ban the sale of new petrol and diesel cars by 2040.

Assess the view that regulation is a better policy for dealing with the problem of air pollution than the allocation of property rights or taxation.

[25 marks]

20.

In 2018, the UK Government will impose a new tax on sugary drinks, aimed at tackling obesity. Behavioural economists believe that taxation alone is unlikely to be very effective in reducing consumption. By 2050, obesity is predicted to affect 60% of adult men, 50% of adult women and 25% of children. It is associated with a range of health problems including type 2 diabetes, cardiovascular disease and cancer.

Explain why, in a free market, sugary drinks may be overconsumed.

[15 marks]

21.

INVESTIGATION

Scenario

You are an economist working for the Ministry of Housing, Communities and Local Government (MHCLG).

You have been asked to investigate the state of the UK housing market.

As part of this investigation, you are to provide answers to three questions.

Referring to the **Extracts A, B, C and D**, use them and your knowledge of economics to help you answer this question.

After considering **Extract E**, and the original evidence in **Extracts A, B, C and D**, would you recommend that the Government should increase its spending on housing? Justify your recommendation.

[25 marks]

22.

According to Public Health England, poor air quality is the largest environmental risk to public health in the UK. In particular, diesel cars are estimated to be responsible for 40 000 premature deaths each year. In response, the Government announced in 2017 that it would ban the sale of new petrol and diesel cars by 2040.

Explain why the use of petrol and diesel cars may be a source of market failure.

[15 marks]

23.

Extract C (lines 17–19) states: 'Many environmental economists suggest that making firms pay for the costs of landfill associated with their products, or taxing raw material inputs, will be necessary to ensure a sustainable future for all.'

Using the extracts and your knowledge of economics, evaluate policies, if any, that governments in developed countries should pursue to reduce waste and encourage the recycling of waste products.

[25 marks]

24.

After considering **Extract E**, and the original evidence in **Extracts A, B, C and D**, which policies, if any, would you recommend that the UK government should introduce to ensure that it achieves its zero-emissions target by 2050. Justify your recommendation.

[25 marks]

25.

Extract B (lines 8–10) states: 'many companies intentionally make it difficult to repair their products in order to increase sales. This is both economically inefficient and environmentally foolish; it imposes costs on the environment even though it may make sense for individual companies.'

With the help of a diagram, explain how the production of goods which are designed not to last long may result in market failure.

[9 marks]